Victor Vimbert-Guerlais

Electronics – embedded systems – FPGA

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github.com/helloHackYnow

Education

CentraleSupélec *2026 – present*

Engineering degree program, electronics track (1st year).

Lycée Sainte-Geneviève, Versailles *2024 – 2026*

Intensive two-year preparatory program (CPGE) for top French engineering schools: MPSI then MP*. Selected to deliver speeches on behalf of the school's student body at major official events (2025 – 2026).

French national baccalauréat *2024*

Majors in mathematics and physics, additional advanced-mathematics program.

Projects

Transport-triggered processor design framework *Amaranth HDL / FPGA, 2026*

Parametrizable transport-triggered architecture written in Amaranth HDL (Python), targeting Alchitry FPGAs. Complete framework enabling rapid creation and iteration of CPU designs. Full design-to-execution flow: macro-capable assembler, C → TTA compiler (LLVM IR canonicalization + list scheduling), simulation/regression framework with *in-hardware* assertions (dedicated test FU), Yosys resource estimation, per-CPU boot ROMs.

2D rendering and audio engine *C++ / CMake / CI, 2025 – 2026*

C++ code linking C graphics and audio libraries; custom 2D renderer (framebuffer, shader pipeline, bloom post-processing), audio engine, ImGui editor, JSON project format and exporter; CMake build with GitHub Actions CI.

UART FPGA ↔ PC debug protocol *Verilog + C/C++, 2025*

Bidirectional labeled-value streaming with remote reset and trigger from the PC; plug-and-play Verilog endpoint module plus a C/C++ client (CMake).

Visual logic-gate circuit simulator *C++ / raylib, 2024*

Interactive circuit simulator: boolean components wired as a directed graph and evaluated by a dedicated simulation engine; custom graphical editor (raylib/raygui) with a component library, selectable curved connections and custom connection colors.

n-body gravity simulator *C++ / ImGui / OpenGL, 2023*

2D n-body gravity simulation with an ImGui interface and a GLSL/OpenGL renderer.

Other

C++ apps for NumWorks calculators (EADK SDK, CI): a ray tracer shipped as an external app; a reusable on-screen textbox component published as a public template. nix-fpga: composable Nix environments for FPGA toolchains (Vivado, open-source EDA); personal NixOS setup (Hyprland, Neovim).

Skills

Hardware / EDA: Amaranth HDL, Verilog, Vivado, Yosys, LiteX (familiar), testbenches and hardware assertions.

Embedded / protocols: UART, ARM GCC, register-level bring-up, bootloaders / boot ROMs.

Programming languages: C, C++, Python, Nix, Lua, assembly (custom ISA).

Tooling: Git, CMake, GitHub Actions, pytest.

CAD: Autodesk Fusion 360.

Spoken languages: French (native), English (B2–C1).

Open-source ecosystems: familiar with the Alchitry and NumWorks ecosystems (tooling, documentation, communities).